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School of Electrical Engineering and Computer Science

IsharaCV

Semester Project

Object Oriented Programing

By

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&

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Problem Statement:

In modern computing devices interaction with humans depends heavily upon input devices which require physical contact, while being effective this is a problem where physical touch is inconvenient and a limitation, or sometimes the users' hands are occupied with another task and multitasking is required

In addition, the use of a third party device feels less direct and lacks natural interaction, which feels less intuitive. And *there is a growing need for a seamless more natural way of communication*.

Solution Proposal:

To address this problem, we have come up with **IsharaCV** – A computer vision based Gesture to input translator which enables humans to communicate with their computer directly eliminating the need for external input devices

IsharaCV uses a real time feedback camera to recognize pre-mapped hand gestures to navigate the computer. It detects a gesture translates it into conventional computer based input and performs the required task

By eliminating the need for input devices **IsharaCV** provides:

- A contactless UI
- Improved accessibility for users with physical limitations
- An improved and intuitive way to communicate with the computer
- Flexibility in environments where conventional input devices aren't practical

Object Oriented Programming usage:

Through this project we've demonstrated how Object Oriented programing can help us solve everyday challenges by finding and implementing new solutions

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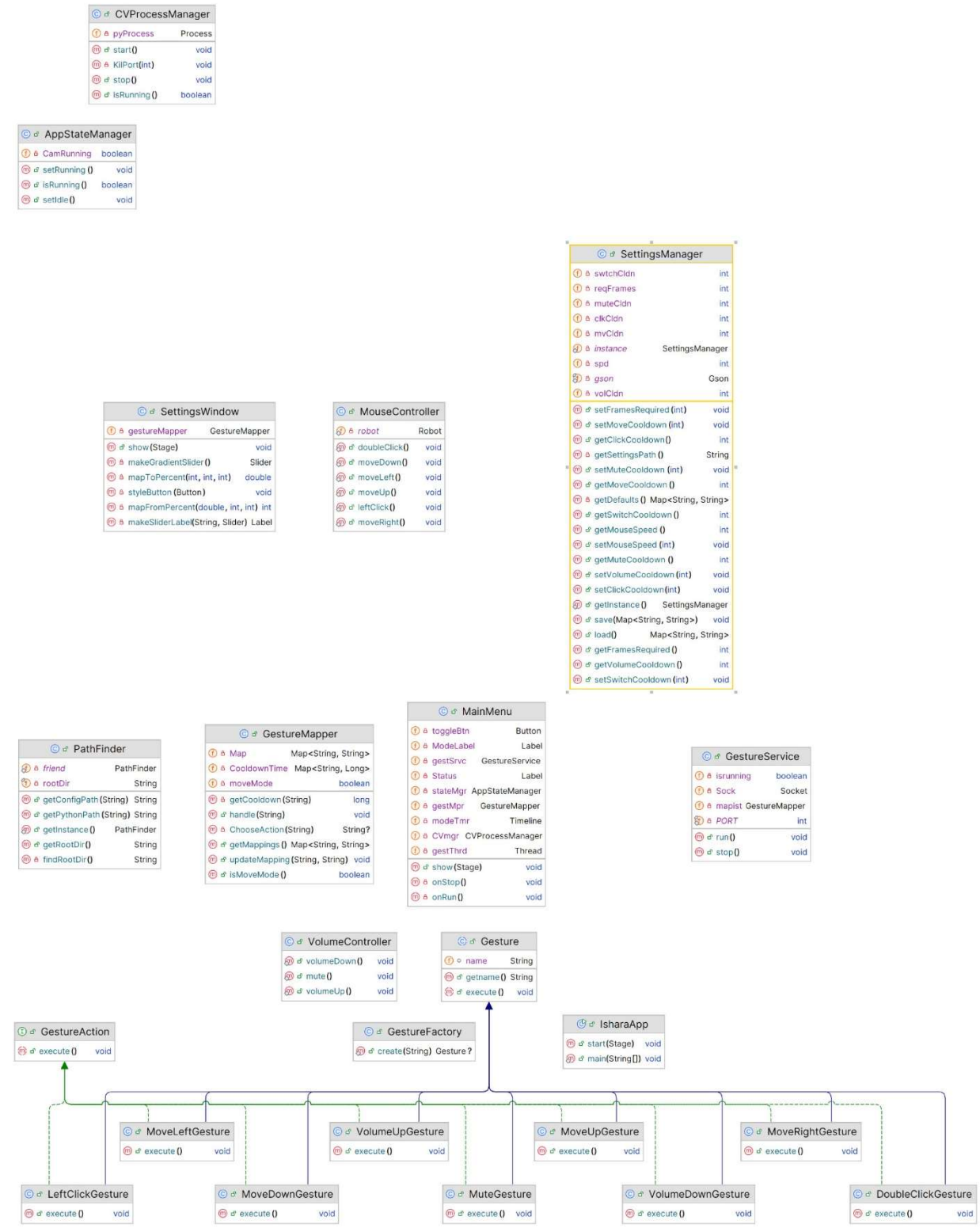
UML



System Architecture:

The App follows a hybrid architecture utilizing both Python and Java to deliver hands-free desktop control.

- Python / mediapipe handles webcam gesture recognition.
- Java utilizes JNA and Robot to carry out desktop actions based on the received signal.
- The Python CV server communicates with the Java app using a TCP socket on the device.
- JavaFx utilized for GUI.

The directory itself is organized in a standardized pattern, first separating Java and Python files, then further dividing Java files into src, lib , and bin folders.

The src folder is further partitioned to separate and group together Java class files based on similarity in functions

Features:

Primary features include:

- Minimalistic and convenient gui.
- Live MainMenu status and mode indicators.
- Start/Stop toggle to run CV engine at will.
- 2 distinct modes, “Volume” and “Move”.
- Precise volume control; Up, Down, Mute.
- 4D mouse movement with the ability to Click and Double Click
- Seamless switch between modes with a gesture
- Intuitive settings window with a Tuning and Instructions Tab
- Detailed tuning options for both modes ie; Mouse speed, Detection sensitivity
- Instructions tab with gestures and their mapped functions for beginners.

Classes used:

Core system classes:

- CVProcessManager
  Responsible for starting and killing Python Engine.
- AppStateManager
  Tracks whether CV engine is running or Idle only.
- SettingsManager
  Manages setting related tasks
- IsharaApp
  The entry point to the app, creates core objects and forwards them to MainMenu.

Communication Layer:

- GestureService
  Receives names strings constantly from the server on a separate thread and forwards them to GestureMapper

Gesture Handling

- Gesture (base/abstract class)
  Abstract class, Parent class of all gestures
- GestureFactory
  Receives action string from Mapper and constructs and returns the corresponding Gesture subclass Object.
- GestureMapper
  Maps received strings from GestureService to appropriate actions while checking cooldowns.

Gestures

- MoveLeftGesture

- MoveRightGesture
- MoveUpGesture
- MoveDownGesture
- LeftClickGesture
- DoubleClickGesture
- VolumeUpGesture
- VolumeDownGesture
- MuteGesture

Action / Interface Layer

- GestureAction
  Interface, defines execute() method for all gestures

Controllers

- MouseController
  Uses Robot class to perform mouse functions.
- VolumeController
  Uses JNA to perform volume related functions.

UI

- MainMenuTest
  First Gui class, primary control centre.
- SettingsWindow
  Second Gui class, contains instructions and tuning controls.

Utility

- Pathfinder
  Finds the app's root directory (IsharaApp) at runtime.

(it is worth noting only classes native to this project are mentioned)

App Workflow:

User side:

User clicks IsharaApp.exe-> Mainmenu opens-> click start-> Python engine starts up

->make gestures to perform functions.

BackEnd:

Startup;

Start pressed ->MainMenu called-> Pathfinder called and finds root directory and returns full path to server-> GestureServer reads json for settings->mediapipe model loaded->Python clears port 6767 and connects-> GestureService is created in a separate thread-> Status changes to running.

Gesture Processing:

Python reads webcam frames at 30fps-> GestureServer checks landmark positions-> If gesture persists for a number of frames its NAME is passed through socket-> GestureService receives NAME, processes it and forwards to mapper->Checks cooldown and maps NAME to corresponding ACTION forwarding to GestureFactory-> GestureFactory creates subclass object->Object calls execute() polymorphically -> corresponding controller method performs desktop action.

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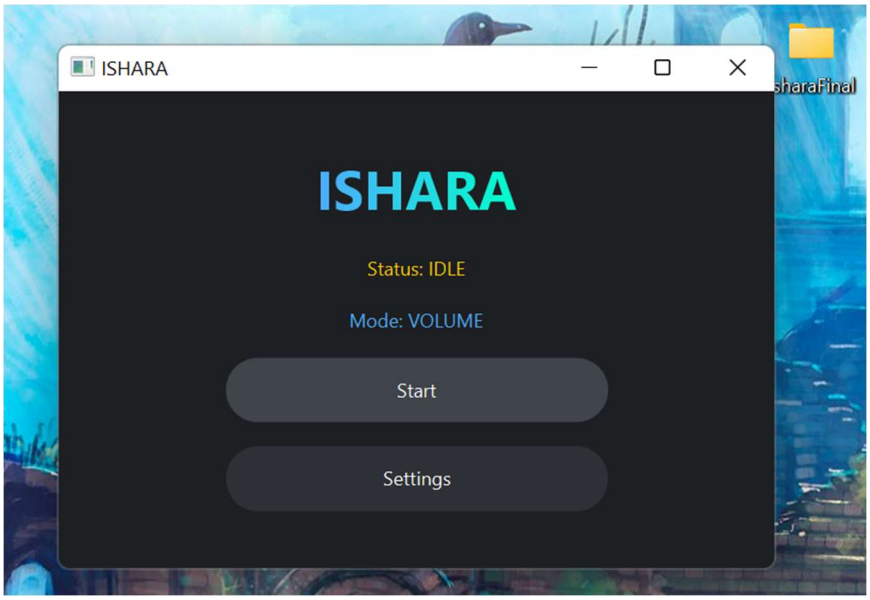
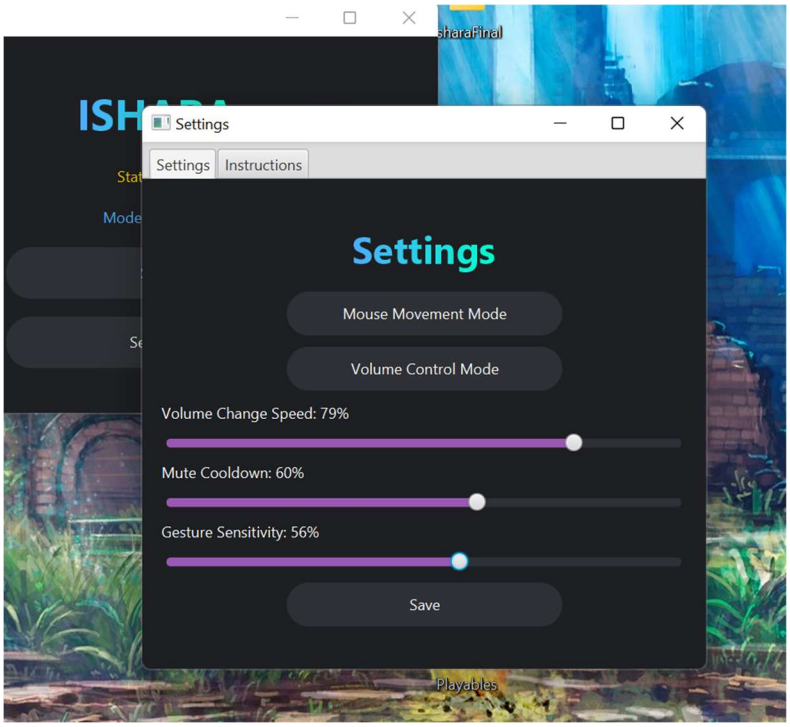
Screen Shots:





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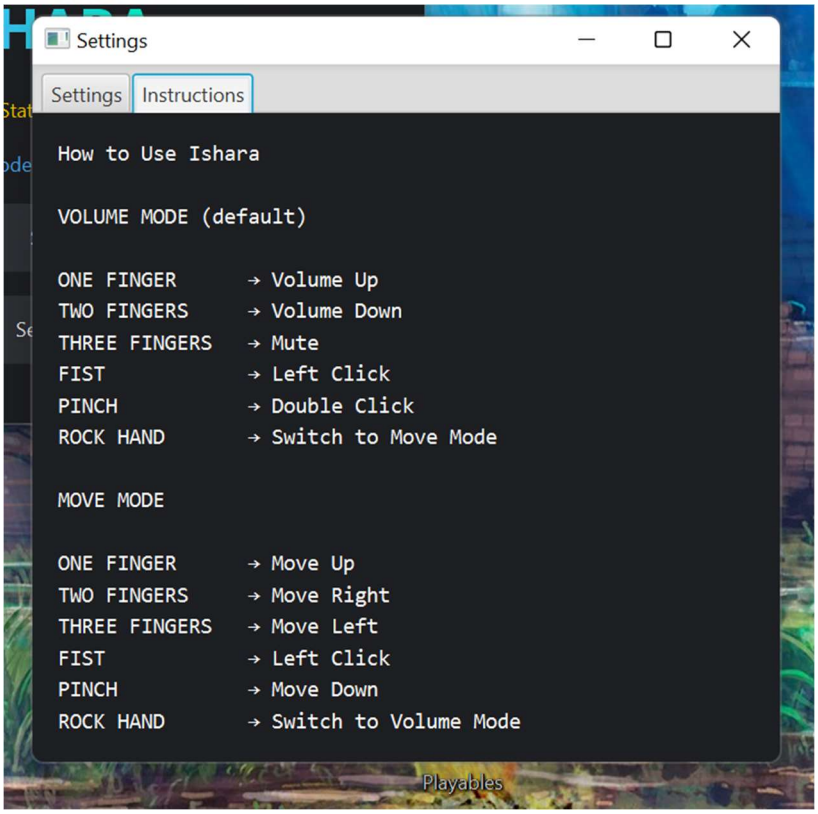

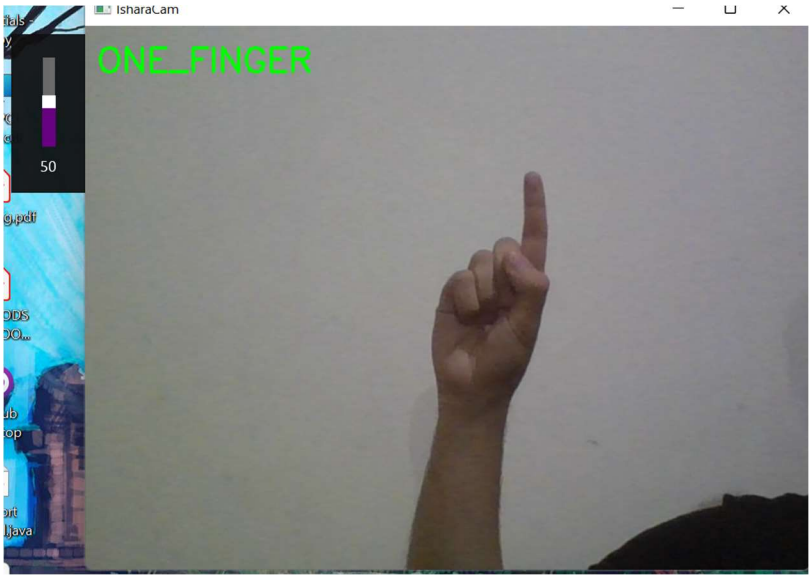

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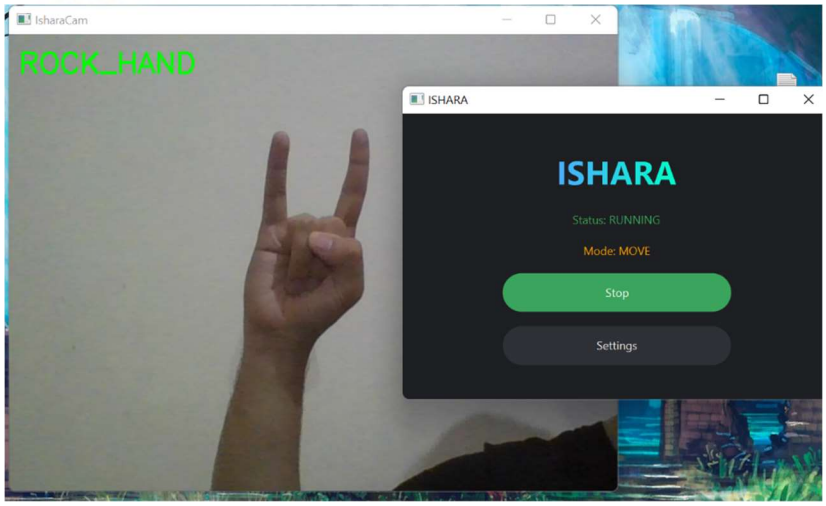

Github Repo Link:

https://github.com/BilalGh0zi/IsharaCV.git

Future Improvements:

- Linux support
- Two handed gestures
- Optional light frame on screen to counter low light scenarios
- More rigorous detection process